

11. Introduction

High-frequency noise appears as sudden random intensity changes in an image due to electronic interference, transmission errors, or sensor noise in the frequency domain this noise is represented by high frequency components.

A Gaussian low pass filter is the most suitable filter because it suppresses

working:-

1. convert the Image into the frequency domain using the fast fouries Transform.

2. Design a Gaussian low pass filter with a suitable cutoff frequency

3. Multiply the filter with the fourier spectrum of the Image

Justification:-

The Gaussian low pass filter provides a smooth transition between passed and blocked frequencies. Unlike the Ideal low pass filter it minimizes.

Application:-

- medical image enhancement
- satellite image processing
- CCTV surveillance
- digital photography

Impact on Image Quality

- Removes high frequency noise
- produces smoother image
- preserves overall brightness and shape

conclusion:-

The Gaussian low pass filter is the preferred frequency-domain filter for reducing

12. Gaussian Low pass filtering:-

Introduction:-

The low-frequency components of an image contain the overall appearance, smooth intensity variations, and major objects. High-frequency components usually represent noise and fine details. Therefore Gaussian low pass filtering is used to preserve the important low frequency

Working:-

1. Transform the image into the frequency domain using FFT
2. Create a Gaussian low pass filter with the desired cutoff frequency
3. Apply the filter to suppress high frequency components

Advantages

- preserves important image information
- Reduce random noise effectively
- produces smooth and natural images

Applications

- Medical Imaging
- Remote sensing
- Image restoration

Result:-

The output image becomes smoother with significantly reduced noise while maintaining

13. High pass filter

Introduction:-

Edges represent rapid intensity changes in an image and are contained in the high frequency components. To enhance these edges, a high pass filter is used because it suppresses low frequency information while emphasizing high frequency details.

Working

1. Convert the image into the frequency domain using FFT
2. Apply a High pass filter
3. Remove low-frequency components
4. Preserve and enhance high-frequency components

Justification

High pass filtering increases image sharpness by highlighting edges, textures, and fine details. It is commonly used in image enhancement and feature

Effect on frequency components

- Low frequencies are suppressed
- High frequencies are enhanced
- Fine textures become clearer

Impact on Image Quality

- Improve sharpness
- Enhance object boundaries

14. Gaussian low pass filtering

Image smoothing is performed to reduce unwanted noise and intensity variations. In the frequency domain, smoothing is achieved by suppressing high-frequency components while working

filters out high-frequency components

- Retains low frequency information representing
- produces gradual intensity transitions
- pixels with intensity less than are classified as the background

• choose a threshold value

Justification:

Effect on frequency components:

- Low frequencies are suppressed
- High frequencies are enhanced
- Fine textures are enhanced
- Impact on Image Quality

15. Suitable Segmentation Technique

Working:

- Choose a threshold value T
- Pixels with Intensity greater than T are classified as the object
- Pixels with Intensity less than T are classified as the background

Justification:

- Very effective when the object is significantly brighter than the background
- Simple, fast, and computationally efficient

Effectiveness:-

- Accurate object extraction
- Easy Implementation
- Best result when there is good contrast
- Accurate object extraction

Application:

- Medical Image enhancement
- Satellite Image processing
- CCTV Surveillance
- Digital photography

Impact on Image Quality

- Remove high frequency noise
- Produces smoother Image
- Preserves overall brightness and shape

Conclusion:-

Suitable segmentation Technique Low pass filter in the prefer frequency domain filter for reducing

Advantages:-

- Preserves Important Image Information
- Reduce random noise effectively
- Produces smooth and natural Images

Application

- Medical Image
- Remote sensing
- Image restoration